

Load-Robust, Physics-Informed Machine Learning for Stator Inter-Turn Fault Diagnosis in Induction Motors: A Leakage-Safe Cascade for Detection, Severity, and Faulted-Phase Identification

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Abstract—Stator winding turn-to-turn (inter-turn) short-circuit faults are a leading cause of induction-motor failure, and early, load-robust diagnosis from non-invasive stator-current measurements remains an open problem. Most reported machine-learning (ML) studies achieve near-perfect in-distribution accuracy but do not test generalization to unseen operating points, and the field rarely releases code or uses leakage-safe evaluation. This paper presents a physics-informed, leakage-safe ML framework that performs hierarchical diagnosis: fault-onset detection, healthy/faulty classification, ordinal severity estimation (percentage of shorted turns), and faulted-phase identification, all from the three-phase stator current. A 357-case PSCAD/EMTDC dataset (sampled at 25 kHz, fundamental 50 Hz) is segmented into 18,033 labelled windows and described by 72 physics-grounded features (extended Park’s-vector severity factor, symmetrical-component ratios and angles, Park ellipse geometry, harmonic ratios). Models are evaluated with a strict group-wise protocol and, crucially, with Leave-One-Load-Out (LOLO) cross-validation. We show that detection (macro-F1 = 0.95) and faulted-phase identification (macro-F1 = 0.997) are intrinsically load-robust because they ride on load-invariant negative-sequence signatures, whereas absolute severity is load-dependent; a SHAP-guided analysis isolates the cause and a load-aware reformulation restores cross-load severity (within-one-level accuracy $0.73 \rightarrow 0.95$). The end-to-end cascade reaches within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection (expected calibration error = 0.021), graceful noise degradation, and 100% onset detection. A 1-D residual network baseline is outperformed by the interpretable feature-based model under limited data. The framework is fully reproducible and openly released.

Index Terms—Induction motor, inter-turn fault, stator winding, fault diagnosis, severity estimation, machine learning, negative-sequence current, leakage-safe evaluation, conformal prediction.

I. Introduction

INDUCTION motors (IMs) drive the majority of industrial processes, and stator winding insulation degradation is among their most common and consequential failure modes. An incipient inter-turn (turn-to-turn) short circuit (ITSC) shorts a small fraction of the turns of one

phase winding; the circulating fault current accelerates local heating and, if undetected, escalates to a phase-to-ground or phase-to-phase fault and catastrophic machine loss [1], [2]. Reliable diagnosis at low severity is therefore of high economic and safety value, and is most attractive when performed non-invasively from the readily available three-phase stator current (motor-current signature analysis, MCSA) [3]. Surveys attribute roughly a third of induction-motor failures to stator winding insulation, and the time between an incipient inter-turn short and a destructive fault can be short, so an on-line monitor must be both sensitive to small faults and trustworthy enough to trigger maintenance without excessive false alarms. Current-based monitoring is preferred over vibration or thermal sensing because the current is already measured by the drive, requires no additional instrumentation, and is robust to mounting and environmental variation.

Two practical questions must be answered for an actionable diagnosis: how severe is the fault (the fraction of shorted turns), and which phase is affected. A large body of recent work applies machine learning (ML) and deep learning (DL) to current-based stator-fault classification with reported accuracies frequently approaching 100% [3], [8], [9], [10]. However, a comprehensive review of supervised ML for IM faults [3] explicitly flags three systemic weaknesses: (i) suspiciously perfect accuracies, often from train/test partitions that allow information leakage; (ii) almost no released code or standardized public datasets; and (iii) very little validation across changing operating conditions. The negative-sequence current is the established physical indicator of inter-turn asymmetry [4], [5], yet its load dependence is rarely confronted in a controlled, reproducible manner.

This paper addresses those gaps. We treat ITSC diagnosis as a hierarchical problem (onset $\rightarrow$ detection $\rightarrow$ severity $\rightarrow$ faulted phase) and build a physics-informed, leakage-safe pipeline whose central methodological device is Leave-One-Load-Out (LOLO) evaluation: training on a subset of mechanical loads and testing on a held-out load. LOLO exposes which diagnostic sub-tasks genuinely generalize and which merely interpolate, and it overturns the optimistic picture given by conventional random/grouped cross-validation. The contributions are:

- 1) A leakage-safe evaluation framework for stator-

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Code and result artifacts are publicly available at <https://github.com/Regal-s/induction-motor-fault-diagnosis>.

fault ML—grouping windows by operating point and reporting LOLO as the primary metric—demonstrating that in-distribution scores substantially overstate cross-load generalization.

- 2) A physics-feature analysis (SHAP attribution plus a Fisher-versus-load-drift study) showing that detection and faulted-phase identification are load-invariant whereas absolute severity is load-dependent, and a load-aware severity reformulation that recovers cross-load within-one-level accuracy from 0.73 to 0.95.
- 3) A complete, openly released hierarchical cascade integrating negative-sequence fault-onset detection, calibrated detection, ordinal severity estimation, and faulted-phase identification, benchmarked honestly—including against a 1-D residual network—under noise and with conformal severity intervals.

II. Related Work

A. Physics-based stator-fault indicators

Penman et al. [1] established the inter-turn sideband frequencies in the stator current, and Cardoso et al. [2] introduced the Extended Park’s Vector Approach (EPVA), in which the $2f_0$ spectral component of the current-vector modulus scales with winding asymmetry. The negative-sequence current and negative-sequence impedance are widely used asymmetry descriptors [4], [5]; phasor-compensation schemes suppress the baseline asymmetry caused by supply voltage unbalance. Mazzeletti et al. [6] use the positive-sequence fifth-harmonic current during soft-starter start-up to distinguish ITSC from high-resistance connections, deriving an analytical severity relation. Singh and Shaik [7] combine the Stockwell transform with support-vector machines (SVMs) to detect, type, and localize stator faults, achieving 100% faulted-phase identification for turn faults at a single severity.

B. Data-driven stator-fault diagnosis

Pietrzak and Wolkiewicz [8] feed short-time-Fourier-transform amplitudes of the symmetrical-component currents to SVM/MLP classifiers for permanent-magnet machines, reporting 96–99% severity classification. Aasar et al. [9] compare ANN, LSTM, and a physics-informed network (PINN) on a public current-plus-flux ITSC dataset, reaching 89% on a seven-level severity task and 99–100% binary detection, and observe that intermediate severities overlap—the hardest case. Ali et al. [10] convert signals to spectrogram images and ensemble several convolutional networks. The review [3] catalogues dozens of such studies and the recurring evaluation and reproducibility shortcomings that the present work targets.

C. Evaluation rigor

The vibration-diagnosis literature has documented how windowing one recording into many overlapping samples

and splitting them at random leaks information and inflates accuracy; group-wise (subject-wise) splitting is the accepted remedy. We adopt group-wise splitting and add LOLO to test genuine cross-condition generalization, which the IM stator-fault literature largely omits.

D. Deep architectures, transfer, and augmentation

Deep models have become standard for current-based diagnosis: one-dimensional convolutional networks operate directly on raw current [16], while two-dimensional networks classify time-frequency images such as short-time-Fourier or wavelet scalograms [10]. Recurrent and temporal-convolutional models [24], attention/Transformer encoders [23], and—most recently—linear-time state-space models [18] and interpretable Kolmogorov–Arnold networks [19] have been explored, although almost exclusively on bearing vibration rather than three-phase stator current. Because labelled fault data are scarce and operating conditions vary, two further lines are active: transfer learning / domain adaptation to bridge load, speed, machine, and simulation-to-experiment gaps, and generative data augmentation (conditional GANs, time-series diffusion) to synthesize rare regimes. A recurring caution, which our normalization respects, is that augmentation or preprocessing applied independently per phase can manufacture a spurious negative-sequence component and thus a fake fault. These observations motivate the architecture, transfer, and augmentation extensions outlined in Section VIII.

III. Dataset and Problem Formulation

A. Simulation dataset

A three-phase squirrel-cage induction motor with stator inter-turn faults was simulated in PSCAD/EMTDC. The fundamental frequency is $f_0 = 50$ Hz and the sampling rate is $f_s = 25$ kHz (500 samples per cycle). Each run is 10 s (250,000 samples). The recorded three-phase stator currents i_a, i_b, i_c (in kA) are used throughout. The fault factor space comprises: severity $\mu \in \{0.3, 0.5, 1, 2, 3, 4, 5\}\%$ of shorted turns; faulted phase $\in \{A, B, C\}$; mechanical load $\in \{\text{NL}, 20, 40, 60, 80, 100\}\%$; and, for an additional subset, the fault inception instant. A healthy set spans loads from 0% to 100%. The corpus contains 357 simulation cases (336 faulty, 21 healthy).

B. Operating timeline

Every faulted run follows a fixed schedule, verified from the per-cycle current envelope: the motor is energized at 2.0 s, the mechanical load is applied at 4.0 s, the inter-turn fault is inserted at 7.2 s (sample 180,000), and it clears about 0.8 s later. Healthy runs are 5 s. Fig. 1 shows the phase-A current and its per-cycle RMS envelope with the annotated events.

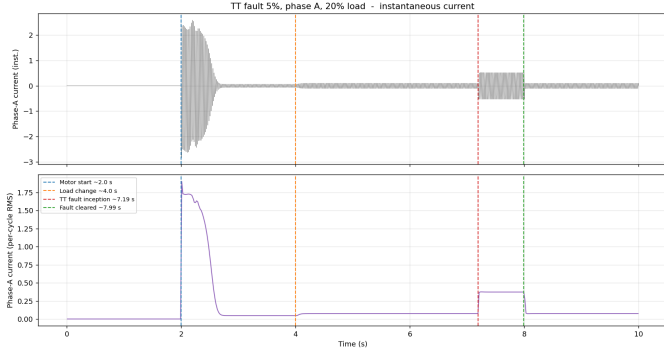


Fig. 1. Phase-A current (top) and per-cycle RMS envelope (bottom) for a faulty case, with motor start (2.0 s), load change (4.0 s), fault inception (7.2 s), and clearing (8.0 s) marked. The fault interval supplies faulty windows; the pre-fault loaded region supplies matched-load healthy windows.

C. Problem statement

Given a short window of three-phase current, the system must (i) decide healthy versus faulty, (ii) if faulty, estimate the severity level μ , and (iii) identify the faulted phase. We also require an online estimate of the fault-onset instant. Because severity is naturally ordered, it is treated as an ordinal target; the operationally meaningful metric is within-one-level accuracy.

IV. Methodology

A. Windowing and matched-load healthy data

Each recording is segmented into windows of $W = 2000$ samples (four cycles). Within the 0.8 s fault interval a 75% overlap is used to compensate for its short duration; the pre-fault steady region is sampled more sparsely. Crucially, the pre-fault loaded region of every faulty run (genuinely healthy, since the fault is inserted only at 7.2 s) is labelled as healthy. This yields matched-load healthy data at the same loads as the faulty windows, preventing the detector from separating classes by load level rather than by fault. Segmentation produces 18,033 labelled windows (11,760 faulty, 6,273 healthy), balanced across severity and phase.

B. Normalization

To remove load-induced amplitude while preserving the inter-phase imbalance that encodes the fault, all three phases of a window are divided by a single per-recording scale. We additionally derive load-invariant magnitude features by dividing each magnitude feature by the within-window positive-sequence fundamental $|I_1|$, which makes the descriptor independent of the operating point; this is central to the load-aware severity result of Section VI-B.

C. Physics-grounded feature extraction

Each window is mapped to 72 features organized in three tiers.

Tier 1 (symmetrical components and EPVA). Per-phase 50 Hz phasors are obtained by a single-bin DFT

TABLE I
Feature groups (72 features per window).

Group	Features
Symmetrical comp.	$ I_0 , I_1 , I_2 $; $ I_2 / I_1 $, $ I_0 / I_1 $; $\angle I_1, \angle I_2$ (+ sin/cos); compensated I_2^c (mag./ratio/angle)
EPVA / Park	severity factor SF; Park-modulus mean/std; ellipse major/minor axes, eccentricity, axis ratio, tilt
Harmonics	third-harmonic ratio and THD, per phase and mean
Per-phase stats	RMS, peak, crest, std, skewness, kurtosis ($\times 3$) and inter-phase differences
Load-invariant	$ I_1 $ -normalized variants of all magnitude features

and combined by the Fortescue transform into positive, negative, and zero sequences I_1, I_2, I_0 . We compute the magnitudes, the ratios $|I_2|/|I_1|$ and $|I_0|/|I_1|$ (the canonical inter-turn severity indicators), the negative-sequence angle $\angle I_2$ and its sine/cosine (the faulted-phase cue), and a per-case compensated negative sequence I_2^c obtained by subtracting the case baseline. The EPVA severity factor is

$$\text{SF} = \frac{A(2f_0)}{\text{DC}} \quad \text{of} \quad |\mathbf{i}_P(t)| = \sqrt{i_\alpha^2 + i_\beta^2}, \quad (1)$$

where i_α, i_β are the Clarke components and $A(2f_0)$ is the amplitude of the 100 Hz component of the Park-vector modulus.

Tier 2 (spectral and geometric). Third-harmonic ratio and total harmonic distortion per phase; Park-vector ellipse descriptors (eccentricity $e = \sqrt{1 - (b/a)^2}$, axis ratio, tilt angle) from the 2×2 covariance of (i_α, i_β) .

Tier 3 (statistics). Per-phase RMS, peak, crest factor, standard deviation, skewness, and kurtosis, together with their inter-phase differences (which directly encode asymmetry), and the $|I_1|$ -normalized variants of all magnitude features.

The Clarke transform underlying the EPVA and the Park ellipse is

$$i_\alpha = \frac{2}{3}(i_a - \frac{1}{2}i_b - \frac{1}{2}i_c), \quad i_\beta = \frac{1}{\sqrt{3}}(i_b - i_c), \quad (2)$$

and the fundamental symmetrical components follow from the Fortescue transform of the per-phase phasors $\mathbf{I}_a, \mathbf{I}_b, \mathbf{I}_c$:

$$\begin{bmatrix} \mathbf{I}_0 \\ \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \quad (3)$$

The negative-sequence ratio $|I_2|/|I_1|$ grows monotonically with μ and is the primary severity descriptor, while $\angle \mathbf{I}_2$ separates into three $\sim 120^\circ$ -spaced clusters that identify the faulted phase. Table I summarizes the feature groups.

Severity is encoded as an ordinal rank $r \in \{0, \dots, 6\}$ over the seven levels $\{0.3, 0.5, 1, 2, 3, 4, 5\}\%$; a gradient-boosted regressor predicts a continuous rank that is rounded and clipped, so that the error metric (mean absolute error in levels and within-one accuracy) respects the natural ordering.

D. Leakage-safe evaluation

Windows are grouped by operating point (severity $\times$ phase $\times$ load); a regular case and all its inception-time variants share a group. Two protocols are used: (a) Stratified Group k -Fold ($k=5$), giving the in-distribution estimate; and (b) Leave-One-Load-Out (LOLO), in which all windows at one load form the test set and the model trains on the remaining loads. LOLO is the primary metric: it measures generalization to an operating point never seen in training and guards against a model that has learned load rather than fault physics.

E. Hierarchical cascade

Stage 0 (onset). A per-cycle negative-sequence fault index $\rho(k) = |I_2(k)|/|I_1(k)|$ is computed and compared against a trailing pre-fault reference window $\mathcal{R}$. An onset is declared at the first cycle for which

$$\rho(k) > \bar{\rho}_{\mathcal{R}} + \kappa \sigma_{\mathcal{R}}, \quad \kappa = 6, \quad (4)$$

holds for three consecutive cycles, where $\bar{\rho}_{\mathcal{R}}$ and $\sigma_{\mathcal{R}}$ are the reference mean and standard deviation. This change-point rule needs no fault labels and provides the online trigger for the downstream stages.

Stages 1–3. Gradient-boosted trees (XGBoost [11]) implement Stage 1 (healthy/faulty, class-weighted), Stage 2 (severity, ordinal regression on the rank index), and Stage 3 (faulted phase, three-class). Stage 1 and Stage 3 use the full feature set; Stage 2 uses the load-invariant subset (Section VI-B). At inference the detector gates the severity and phase stages: windows predicted faulty are passed to Stages 2 and 3.

Calibration and uncertainty. Stage-1 probabilities are assessed with the expected calibration error (ECE) and Brier score [15]. Severity uncertainty is quantified with conformalized quantile regression (CQR) [13], [14]: lower/upper quantile regressors $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$ are fit on a proper-training split, a conformity score $E_i = \max\{\hat{q}_{\alpha/2}(x_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(x_i)\}$ is evaluated on a disjoint calibration split, and the prediction interval $[\hat{q}_{\alpha/2}(x) - Q, \hat{q}_{1-\alpha/2}(x) + Q]$ uses the finite-sample-corrected quantile Q of $\{E_i\}$, giving distribution-free marginal coverage $1 - \alpha$.

F. Interpretability

SHAP attribution [12] is used both to explain predictions and as a leakage audit: the top features must be physically meaningful (negative-sequence ratio, EPVA factor, Park eccentricity) rather than proxies for load.

V. Experimental Setup

Features are computed in Python (NumPy/Pandas) and models trained with scikit-learn and XGBoost (400–500 trees, depth 5, learning rate 0.05, subsample/colsample 0.8); the deep baseline uses PyTorch. The 1-D residual network (ResNet-1D) baseline has a wide-kernel stem and three residual stages over decimated three-phase windows.

TABLE II
Per-stage performance (mean over folds).

Stage	Metric	Group k -Fold	LOLO
1 Detection	macro-F1	0.977	0.948
	ROC-AUC	0.994	—
2 Severity (load-aware)	within-1	0.997	0.946
	QWK	0.987	0.736 [†]
3 Phase ID	macro-F1	0.972	0.997

[†]QWK is pooled across all held-out loads.

Stage-3 phase (GroupKFold OOF)

	A	B	C
A	3622	298	0
B	35	3885	0
C	0	0	3920
	A	B	C

pred

Fig. 2. Faulted-phase confusion matrix (group k -fold, out-of-fold). Phase C is perfect; only minor A $\leftrightarrow$ B confusion remains. Cross-load (LOLO) phase accuracy is 0.997.

All models share the splits of Section IV-D. Reported metrics are accuracy, macro-F1, severity within-one-level accuracy, mean absolute error in severity levels, quadratic-weighted kappa (QWK), and per-phase F1, computed on out-of-fold predictions and reported as the mean over folds. The detector is class-weighted to compensate for the healthy/faulty window ratio. Stages 2 and 3 are trained only on faulty windows but evaluated end-to-end on the windows the detector predicts as faulty, so the reported cascade numbers reflect realistic error propagation rather than oracle gating. Predictions are additionally aggregated per recording (median severity rank, majority-vote phase), which is the granularity at which a maintenance decision is taken. Feature scalars are fit on the training partition only, and all randomized components use fixed seeds for reproducibility.

VI. Results and Discussion

A. Per-stage performance

Table II reports each stage under both protocols. In-distribution, all stages exceed 0.97. Under LOLO, detection (macro-F1 0.948) and faulted-phase identification (macro-F1 0.997) remain strong, while load-blind severity collapses at unseen light loads. The faulted-phase confusion matrix (Fig. 2) is near-diagonal.

The per-held-out-load LOLO breakdown (Table III) localizes the difficulty: detection is perfect at the four heavier loads but degrades at the no-load (NL) and 20% extremes, where the small fault current of a low-severity ITSC at light load most overlaps the healthy distribution

TABLE III
Per-held-out-load detection (LOLO, macro-F1).

Hold-out load	NL	20%	40%	60%	80%	100%
Detection F1	0.885	0.749	1.000	1.000	1.000	1.000

TABLE IV
Severity (within-one-level) under feature choice.

Severity feature set	Group k -Fold	LOLO
All features (magnitudes incl.)	0.993	0.887
Load-invariant only	0.946	0.702
Load-invariant + $ I_1 $ -norm	0.990	0.729
+ measured load (proposed)	0.997	0.946

at an unseen operating point. This is the practically important incipient-at-light-load regime, and it is visible only under LOLO—random or grouped cross-validation hides it entirely.

B. Why the sub-tasks differ: a load-robustness analysis

SHAP attribution shows that detection and phase identification are driven by the negative-sequence ratio and angle and by inter-phase RMS differences—quantities that are ratios and therefore load-invariant. Phase identification is so robust that its LOLO score exceeds its in-distribution score, because the faulted-phase signature (the $\sim 120^\circ$ -separated negative-sequence angle) does not change with load while more training loads help. In contrast, an initial severity model that used absolute magnitude features collapsed at the no-load extreme under LOLO (within-one-level 0.39). A Fisher-versus-load-drift study confirmed that magnitude features drift strongly across load whereas ratio and $|I_1|$ -normalized features are stable. Restricting the severity stage to load-invariant features and supplying the measured load (a sensor input, not a label) raises LOLO within-one-level accuracy to 0.95 (Table IV). This is the paper’s key technical result: severity is recoverable across operating points once the model is made load-aware.

The error structure (Fig. 3) is band-diagonal: misclassifications occur almost exclusively between adjacent severity levels, so a one-level tolerance captures operationally-equivalent predictions and the quadratic-weighted kappa—which penalizes distant errors more—is high (0.987).

C. End-to-end cascade

With the load-aware severity stage, the gated cascade achieves the end-to-end results of Table V, scoring a window correct if healthy windows are detected as healthy and faulty windows have the correct phase and a severity within one level. Per-recording aggregation (median severity, majority-vote phase) gives within-one-level accuracy of 0.927 in-distribution and 0.918 across unseen loads. Fig. 4 summarizes the four headline metrics for both protocols.

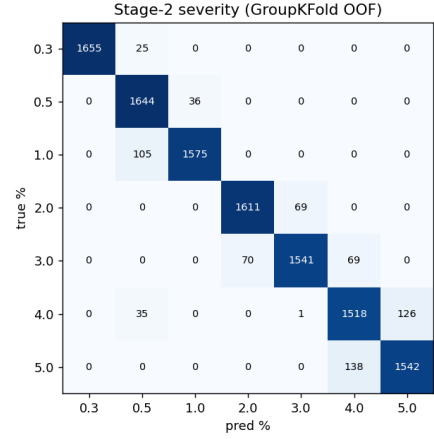


Fig. 3. Severity confusion matrix (group k -fold, out-of-fold, load-aware). Errors are almost entirely between adjacent severity levels, justifying the ordinal treatment and the within-one-level metric.

TABLE V
End-to-end cascade (window level).

Quantity	Group k -Fold	LOLO
Detection macro-F1	0.977	0.948
Severity within-1 (cond.)	0.997	0.945
Phase accuracy (cond.)	0.957	0.997
End-to-end within-1	0.949	0.916
End-to-end exact	0.911	0.521
Per-recording within-1	0.927	0.918

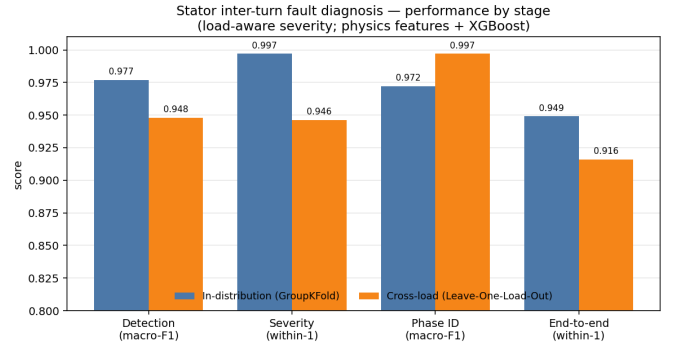


Fig. 4. Per-stage and end-to-end performance, in-distribution (group k -fold) versus cross-load (LOLO). Phase identification is the only metric for which cross-load exceeds in-distribution, reflecting its load-invariance.

D. Noise robustness

Additive Gaussian noise was injected and features recomputed (Table VI). Severity (within-one-level) and phase accuracy are essentially flat down to 20 dB SNR, whereas detection is the noise-sensitive stage—consistent with the physics, since separating a small fault signature from healthy operation is hardest when the signal is buried in noise.

E. Calibration, uncertainty, and onset

Stage-1 probabilities are well calibrated (ECE = 0.021, Brier = 0.021). Conformalized quantile regression for

TABLE VI
Noise robustness (train clean, test noisy; group k -fold).

SNR	Detect macro-F1	Severity within-1	Phase acc.
clean	0.977	0.986	0.958
40 dB	0.927	0.980	0.954
30 dB	0.854	0.975	0.945
20 dB	0.792	0.971	0.939

TABLE VII
Stage-0 fault-onset detection (35 sampled cases).

Metric	Value
Detection rate	100%
Pre-fault false-alarm rate	0%
Median detection delay	0 ms (within first faulted cycle)
Delay at $\mu \geq 0.5\%$	0 ms
Delay at $\mu = 0.3\%$	≈ 1.42 s

TABLE VIII
Feature-based cascade vs. deep baseline.

Task / metric	XGBoost (features)	ResNet-1D (raw)
Detection F1 (group k -fold)	0.977	0.898
Detection F1 (LOLO, load 20%)	0.749	0.401
Phase macro-F1 (group k -fold)	0.972	0.848
Severity within-1 (group k -fold)	0.990	1.000

severity improves interval coverage from 0.49 (naive quantiles) toward the nominal target; we note that the operating-point group structure breaks exchangeability, and group-conditional (Mondrian) conformal prediction is the appropriate remedy. The Stage-0 onset detector flags 100% of faults with no pre-fault false alarms; for severity $\geq 0.5\%$ the fault is detected within the first faulted cycle, while the smallest 0.3% faults incur about 1.4 s of delay because their negative-sequence rise is small (Table VII).

F. Comparison with a deep baseline

On a grouped split the feature-based XGBoost cascade outperforms the ResNet-1D baseline on detection and phase identification, while the two are comparable on severity within-one-level (Table VIII). Under LOLO the gap widens markedly (detection 0.948 vs. 0.401). This is consistent with the literature for limited datasets: physically engineered features with strong inductive bias are more data-efficient and generalize better than a raw-signal network trained on a few hundred distinct operating points. The comparison should be read as a data-efficiency result rather than a claim that deep models cannot help with more data and tuning; richer architectures (state-space and attention models) are part of our ongoing work.

G. Computational complexity and deployment

The feature pipeline is lightweight and causal once trained. Each window requires one real FFT, a Fortescue projection, a Clarke transform with a covariance eigen-decomposition, and a small set of statistics—all

$O(W \log W)$ in the window length W . The gradient-boosted stages have millisecond-scale inference and a memory footprint of a few megabytes, so the full onset→detection→severity→phase chain can run on an embedded edge controller alongside the drive, in contrast to image-plus-CNN pipelines whose spectrogram generation and convolutional inference dominate cost. Because the diagnostic features are interpretable physical quantities, predictions can be audited on-line (e.g., by reporting the negative-sequence ratio and the EPVA factor that drove a decision), which is valuable for safety-critical acceptance.

H. Discussion

The study reframes the common “~100% accuracy” result: such numbers are attainable in-distribution, but only LOLO reveals what truly generalizes. Detection and faulted-phase identification generalize because they rest on load-invariant negative-sequence quantities; absolute severity does not, until the model is made load-aware. SHAP confirms the models use inter-turn fault physics rather than dataset shortcuts, and the released code and result artifacts directly address the reproducibility gap noted in [3].

Several reported figures (phase identification at 0.997, detection near unity at heavier loads) are high enough to warrant scrutiny. We attribute them partly to the cleanliness of simulation, which lacks the sensor noise, mechanical eccentricity, and supply distortion of a real drive; the noise study (Table VI) shows that detection in particular degrades substantially as the signal-to-noise ratio falls, the expected and reassuring behavior. Equally important, the very high phase-identification score is physically defensible: the faulted phase is encoded in the angle of the negative-sequence current, an essentially load- and noise-robust quantity, so the result is not an artifact of a favorable split. By contrast, the gap between in-distribution and LOLO severity quantifies how much a conventional evaluation would have over-promised, and makes the load-aware reformulation a measurable contribution rather than a tuning trick. We therefore advocate reporting cross-condition (LOLO-style) metrics by default, alongside interpretability evidence, so that stator-fault ML results become comparable and trustworthy.

VII. Limitations and Future Work

The principal limitation is that the reported results are from high-fidelity simulation; although the negative-sequence physics that the model relies on is machine-independent, experimental validation is required, and our per-recording normalization currently assumes the availability of a healthy reference segment. To address these points and to advance the method beyond an application of existing tools, we are pursuing five extensions that the recent literature identifies as genuinely open for three-phase stator current:

(i) Sequence-component Stockwell tensor. Rather than imaging raw phase currents, we apply the Stockwell transform to the positive-, negative-, and zero-sequence streams and stack the three time-frequency maps as a tensor input. The negative-sequence channel carries the load-invariant inter-turn signature, and the frequency-dependent resolution of the Stockwell transform tracks the $2f_0$ EPVA line and slip sidebands better than a fixed-window spectrogram.

(ii) Load-invariance-regularized feature selection. We extend the minimum-redundancy–maximum-relevance criterion with a term that penalizes features whose class-conditional statistics drift across load, $J(f) = \text{Rel}(f) - \lambda \text{Red}(f) - \gamma \text{LoadVar}(f)$, yielding a sparse, physically traceable, load-stable subset—built into the selection criterion rather than relegated to an opaque adversarial back-end.

(iii) Phase-coupled, operating-point-conditioned network. A linear-time selective state-space backbone [18] processes the current; a small inter-phase coupling module encodes the asymmetry that defines the fault; feature-wise modulation [20] injects the measured operating point (so the model adapts to load instead of erasing it); and an interpretable Kolmogorov–Arnold head [19] regresses severity, with a physics-informed residual tying it to the sequence-component relation.

(iv) Physics-anchored domain adaptation. Training on simulation and adapting to experiment with alignment anchored on physics invariants (the negative-sequence ratio and unbalance coefficients), validated on the public three-phase ITSC current dataset [21] and the current-plus-flux dataset [22], and across machines.

(v) Physics-constrained generative augmentation. A conditional generator of three-phase current that enforces Kirchhoff current balance as a hard constraint and the negative-sequence–severity relation as a soft constraint, enabling synthesis of unseen load $\times$ severity cells—unlike per-phase augmentation, which would fabricate a spurious imbalance.

We further plan to simulate supply voltage unbalance—the classic confounder that mimics the inter-turn signature—to verify rejection via the voltage-sequence features, to add group-conditional (Mondrian) conformal calibration of the severity intervals, and to broaden the benchmark to a dozen-plus classical and deep models under the same leakage-safe protocol.

VIII. Conclusion

We presented a physics-informed, leakage-safe machine-learning framework for stator inter-turn fault diagnosis in induction motors that performs onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification from the three-phase stator current. By adopting Leave-One-Load-Out evaluation we demonstrated that detection and faulted-phase identification are intrinsically load-robust, while absolute severity is load-dependent and is recovered to within-one-level accuracy of 0.95 across unseen loads once the model is

made load-aware. The end-to-end cascade attains within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection, graceful noise degradation, and reliable onset detection, and it is fully reproducible. The results argue for cross-condition evaluation as standard practice in motor-fault ML and provide an interpretable, openly released baseline for future work.

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